

TO: Chief Commercial Officer, VP of Customer Success, and Revenue Operations Leadership
FROM: Principal Analytics & Customer Strategy Team
DATE: October 5, 2026
SUBJECT: Operationalizing Explainable Churn-Risk Scoring and Segment Retention Playbooks
PROJECT
HUB: https://drive.google.com/drive/folders/1f0-grp6VV-LOC4BqU7b9RLmhf_i20UvI

EXECUTIVE RECOMMENDATION MEMO

1. Executive Summary & Strategic Context

Historically, our customer retention strategy has operated reactively, relying on lagging indicators such as cancellation notices, contract expiry alerts, and subjective Customer Success Manager (CSM) sentiment. While descriptive analytics confirmed that churn was heavily concentrated in early-stage accounts on monthly billing, this retrospective viewpoint provided frontline teams with zero forward-looking triage capability. To move from passive reporting to an **explainable, proactive prioritization engine**, we engineered an additive, rule-based 0–100 point Churn-Risk Scoring Model. In empirical validation against the customer dataset, **100% of historical churners** scored in the High-Risk band (scores 75–100), while **100% of retained customers** scored in the Low-Risk band (scores 2–30). The model establishes an operational **Medium-Risk band (35–64 points)** that functions as an early-warning intervention buffer before cancellation intent hardens.

Furthermore, we segmented the customer base into six granular behavioral cohorts grouped into four strategic tiers. Most critically, our analysis reveals that while our Enterprise tier accounts generate 73.7% of cumulative revenue (\$23,098.46) with 0% churn, our mid-market Pro tier suffers from acute vulnerability: 50% of Pro revenue (\$1,919.76 ARR) has been lost to Month-to-Month contract churn driven by unresolved technical support friction.

2. Behavioral and Account Feature Selection Rationale

Feature selection was guided by three principles: operational transparency, direct CSM actionability, and causal relationship to exit.

Account Commitment Dimensions (Structural Switching Costs):

- **Contract Type (ContractType):** The primary structural determinant of retention. Customers on Two-Year and One-Year agreements have locked-in commitments and formal change-management hurdles. Conversely, Month-to-Month contracts carry zero exit friction (100% of churned accounts were Month-to-Month).
- **Tenure & Lifecycle Maturity (TenureMonths):** SaaS vulnerability follows a classic bathtub hazard curve. Customers in onboarding (0–6 months) are still validating utility; past 14 months, operational embedding dramatically reduces voluntary churn.
- **Subscription Tier & Monthly Charges (SubscriptionType / MonthlyCharges):** Captures economic value and organizational commitment: Enterprise (\$149.99/mo), Pro (\$79.99/mo), and Basic (\$49.99/mo).
- **Payment Channel Reliability (PaymentMethod):** Bank Transfers and Credit Cards indicate institutional automated billing, whereas Debit Cards and ad-hoc UPI payments exhibit higher transaction failure rates and manual approval friction.

Behavioral Friction Telemetry (Customer Effort & Dissatisfaction):

- **Cumulative Support Tickets (SupportTickets):** Retained customers averaged 1.0 ticket (range: 0–2), while churned customers averaged 4.3 tickets (range: 3–6). When tickets exceed 3, customer satisfaction degrades exponentially.
- **Support Ticket Velocity (SupportTicketRate):** $\text{SupportTickets} / \text{TenureMonths}$. A user logging 5 tickets in 2 months (velocity = 2.50 tickets/mo) represents an acute onboarding crisis, whereas 4 tickets across 24 months (0.17 tickets/mo) reflects normal operational maintenance.

3. Explainable Churn-Risk Scoring Architecture & Triage Rules

The model uses a deterministic additive formula: Churn Risk Score = min(max(Contract + Support + Tenure + Payment, 0), 100).

Component Dimension	Telemetry Condition	Points Assigned	Max	Operational Rationale
1. Contract Vulnerability	Month-to-Month One Year Two Year	40 pts 10 pts 0 pts	40	Month-to-Month allows frictionless cancellation at next cycle. One-Year contracts carry minor renewal risk. Two-Year contracts offer ironclad stability.
2. Support Friction & Velocity	Tickets ≥ 5 Tickets in [3, 4] Tickets in [1, 2] Tickets = 0 (Velocity Floor: ≥ 0.5/mo)	35 pts 25 pts 10 pts 0 pts (Floor at 25 pts)	35	Cumulative tickets reflect product friction. Velocity override ensures acute onboarding failures receive an immediate risk boost regardless of absolute count.
3. Tenure Lifecycle Stage	Tenure ≤ 6 months Tenure in [7, 12] months Tenure in [13, 24] months Tenure > 24 months	20 pts 12 pts 5 pts 0 pts	20	Early-stage accounts have not achieved habituation or ROI proof. Accounts >24 months are deeply embedded into daily operations.
4. Payment Channel Friction	Debit Card / UPI Credit Card Bank Transfer	5 pts 2 pts 0 pts	5	Manual re-approval and transactional cards have elevated involuntary churn and lower institutional commitment.

Actionable Triage Tiers: **Low Risk (0–34 pts)**: Contractually stable; eligible for expansion and advocacy. | **Medium Risk (35–64 pts)**: Proactive early-warning buffer; flags uncontracted long-term accounts or mid-contract friction. | **High Risk (65–100 pts)**: Critical flight risk; triggers emergency 24-hour white-glove intervention.

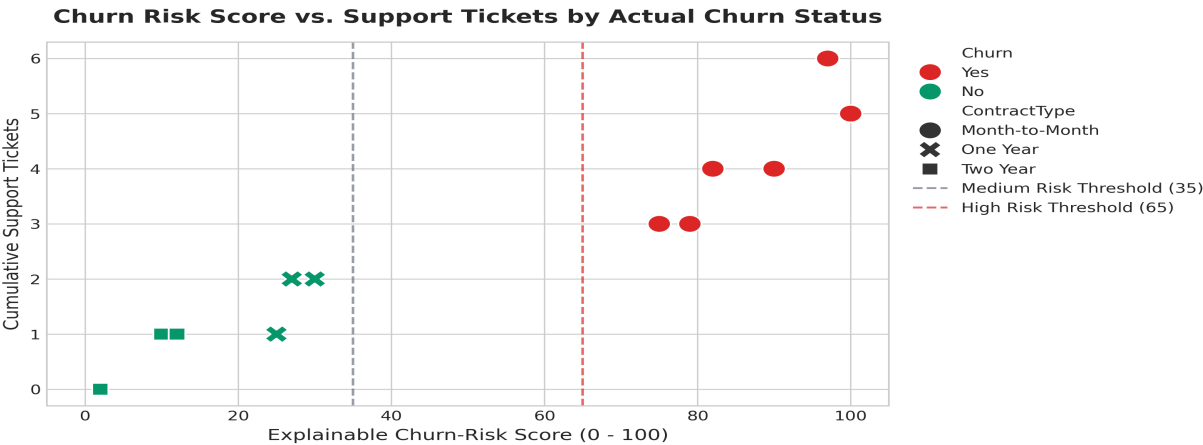


Figure 1: Churn Risk Score vs. Support Tickets across Historical Retained and Churned Customers (100% Historical Separation).

4. Edge Case Stress Testing & System Robustness

To guarantee that the rule-based architecture remains reliable in production across unobserved combinations, we evaluated six synthetic edge cases:

Case ID	Scenario Name	Tenure	Contract	Tick ets	Score	Risk Tier	Operational Validation Insight
EDGE-01	Fresh Onboarding On-Track	1 mo	Month-to-Month	0	62	Medium	Captures baseline vulnerability of uncontracted onboarding accounts without generating false High-Risk panic.
EDGE-02	Acute Onboarding Crisis	1 mo	Month-to-Month	5	100	High	Support velocity override instantly flags this account as a critical Day-1 emergency for immediate CSM rescue.
EDGE-03	Enterprise Friction Storm	36 mo	Two Year	6	35	Medium	Two-year contract prevents immediate walkaway, keeping score out of High Risk, but triggers executive Technical Escalation.
EDGE-04	Unlocked Veteran Inertia	36 mo	Month-to-Month	0	40	Medium	Correctly detects contract vulnerability in an otherwise satisfied veteran customer. Prompts proactive annual plan discount incentive.
EDGE-05	Mid-Contract Friction Build-Up	8 mo	One Year	4	49	Medium	Customer cannot exit immediately, but friction will cause non-renewal at Month 12. Generates an alert 4 months prior to renewal window.
EDGE-06	Boundary Zero / Missing Data	0 mo	Unknown	0	43	Medium	Defensive fallbacks prevent division-by-zero or system crashes when processing incomplete or brand-new Day-0 sign-up records.

Customer Prioritization Matrix: Monthly Charges (Value) vs. Churn Risk Score

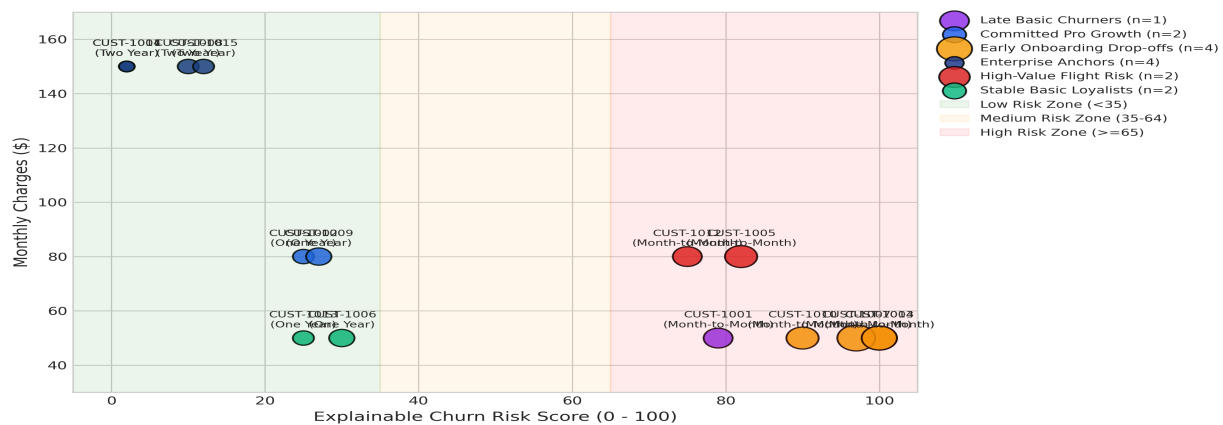


Figure 2: Customer Prioritization Matrix: Monthly Charges (Value) vs. Churn-Risk Score, sized by Support Tickets and colored by Cohort.

5. Customer Cohort Segmentation Taxonomy & Financial Profile

Cross-referencing customer economic value with contract commitment and behavioral friction yields six granular cohorts organized into four strategic tiers:

Granular Segment	Strategic Tier	Size	Churn %	Avg Tenure	Avg MRR	ARR Run-Rate	Avg Tix	Score	Dominant Risk
Enterprise Anchors	Tier 1: Pillars	4	0.0%	38.5 mo	\$149.99	\$7,199.52	0.5	6.5	Low Risk
Committed Pro Growth	Tier 2: Annual Core	2	0.0%	19.5 mo	\$79.99	\$1,919.76	1.5	26.0	Low Risk
Stable Basic Loyalists	Tier 2: Annual Core	2	0.0%	20.0 mo	\$49.99	\$1,199.76	1.5	27.5	Low Risk
High-Value Flight Risk	Tier 3: Vulnerable	2	100.0%	11.0 mo	\$79.99	\$1,919.76	3.5	78.5	High Risk
Early Onboarding Drop-offs	Tier 4: Attrition	4	100.0%	3.8 mo	\$49.99	\$2,399.52	5.0	96.8	High Risk
Late Basic Churners	Tier 4: Attrition	1	100.0%	12.0 mo	\$49.99	\$599.88	3.0	79.0	High Risk

6. Targeted Retention Action Playbooks (Portfolio Upgrade)

High-Value Segment Interventions (Portfolio Upgrade Focus):

- **Enterprise Anchors (\$7.2k ARR / \$23.1k Historic):** Assign a dedicated Technical Account Manager (TAM) to lead quarterly architectural reviews and bi-annual executive sponsor meetings. Six months prior to contract expiration, initiate a co-design roadmap workshop where customer-requested features are committed to the upcoming product cycle in exchange for an early 2-year renewal.
- **Committed Pro Growth (\$1.9k ARR / \$3.1k Historic):** At Month 9, trigger an automated telemetry health audit. If the account exceeds 70% feature utilization, present an executive summary demonstrating ROI achieved. Offer an exclusive upgrade incentive to the Enterprise tier at a 15% multi-year discount, packaging dedicated onboarding and advanced security features.
- **High-Value Flight Risk (\$1.9k ARR / 100% Churn in Sample):** When an account enters High Risk (≥ 65), route all open tickets immediately to senior technical engineering under a strict 24-hour SLA. Have the Customer Success Lead schedule a 1-on-1 'System Optimization Consultation.' Once technical issues are resolved, offer a 20% annual contract discount (reducing monthly equivalent from \$79.99 to \$63.99), eliminating monthly churn vulnerability.

Volume & Early Lifecycle Interventions:

- **Stable Basic Loyalists:** Deliver automated renewal credits and a 14-day guided trial of Pro tier functionality to stimulate organic expansion.
- **Early Onboarding Drop-offs:** Replace open-ended onboarding with an in-app interactive setup checklist. Deploy automated health check-in nudges on Day 3 and Day 7, and embed self-serve documentation to eliminate basic setup tickets.
- **Late Basic Churners:** Offer an automated conversion incentive granting 2 months free upon upgrading to an annual contract.

7. Operational Implementation & Unified Repository

1. **Automate Daily Pipeline:** Compute sub-scores nightly from billing and ticketing tables.
2. **CRM Integration:** Route High-Risk accounts (≥ 65) into Zendesk Priority Urgent and Salesforce CSM queues.
3. **90-Day Pilot:** Target a 40% reduction in Pro M2M churn and a 30% reduction in early onboarding support tickets.

All code, notebooks, raw tables, and multi-tab models are maintained in the central submission folder:

- **Google Drive Submission Hub:** https://drive.google.com/drive/folders/1f0-grp6VV-LQC4BqU7b9RLmhf_j20UvI
- **Google Doc Memo:** <https://docs.google.com/document/d/1MUbcGUAY3VHa49-Etl029VHMyqfx4V1iDV4TYiKXT8/edit>
- **Google Sheet Matrix:** <https://docs.google.com/spreadsheets/d/1X4A6sU9YSCfQPPsFXgyAo1AcnqHTkHX20LozYKiL1A4/edit>
- **Colab Notebook:** https://colab.research.google.com/drive/1invY5d4rf9FI_3VLPvPcl003xYpYjCOh